

SUNIT KUMAR

Senior Azure DevOps & Cloud Engineer

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LinkedIn Profile: [linkedin.com/in/sunit-kumar-63582114](https://www.linkedin.com/in/sunit-kumar-63582114) | **Live DevOps Portfolio:** homepage.sunitdevops.site

Profile

- **Experience:** 16+ years of IT experience, including 7+ years in Azure DevOps and cloud engineering and approximately 9 years in enterprise .NET application development.
- **CI/CD:** Implemented multi-environment CI/CD pipelines using Azure DevOps and GitHub Actions. Migrated pipelines from Jenkins to GitHub Actions and configured self-hosted agents and runners.
- **Cloud and Infrastructure as Code:** Worked with Azure PaaS services and used Terraform and ARM Templates for modular infrastructure provisioning, remote state management and environment-specific configurations.
- **Containers and Kubernetes:** Containerized applications using Docker, stored versioned images in ACR, and deployed workloads to self-managed Kubernetes clusters and AKS using Kustomize.
- **Automation and Security:** Used PowerShell, PowerShell Remoting and Microsoft Graph API to automate deployments, IIS operations, Azure SQL migration and Entra ID administration. Implemented secure access using Workload Identity Federation, HashiCorp Vault, Kubernetes Secrets and TLS.
- **Monitoring and Logging:** Implemented monitoring and centralized logging using Application Insights, Prometheus, Grafana, Fluent Bit, Elasticsearch and Kibana.
- **Key Achievements:** Automated the migration of an Azure SQL database of approximately 250 GB, reducing more than 15 hours of manual effort, and resolved a critical production issue that prevented sample reports from being delivered to customers.

Education

- **Master of Computer Application (MCA)** from Vellore Institute of Technology, Vellore, 2005-2008

Technical Expertise

Primary Expertise (DevOps, Cloud and Automation):

Azure DevOps, Azure Pipelines, GitHub Actions, Jenkins, Git, Terraform, ARM Templates, Docker, Kubernetes, AKS, kubeadm, Kustomize, ACR, PowerShell, PowerShell Remoting.

Azure Services:

Azure App Service, Azure Functions, Azure Storage, Azure SQL, Azure Service Bus, Azure Key Vault, Microsoft Entra ID, Application Insights

Security and Observability: Workload Identity Federation, HashiCorp Vault, Kubernetes Secrets, NGINX Ingress, TLS/HTTPS, Prometheus, Grafana, Fluent Bit, Elasticsearch, Kibana

Secondary Skills (.NET Full Stack Development):

C#, .NET Framework, .NET Core, .NET 6, ASP.NET MVC/Web API, SQL Server, Entity Framework

Employment History

- Encora Innovation Labs | Senior Azure DevOps Engineer | Dec 2023 – Sep 2025 (GoDaddy Client)
- T-Systems ICT India Pvt. Ltd. | Azure DevOps Engineer | Dec 2018 – May 2023 (Royal Dutch Shell - Client)
- NTT Data Services | Senior Software Engineer | Nov 2016 – Mar 2018
- Johnson Controls Pvt. Ltd. | Senior Software Engineer | Jan 2014 – Oct 2016
- Standard Chartered Scope International Pvt. Ltd. | Product Analyst | Sep 2011 – Dec 2013
- Pellucid Healthcare Networks Pvt. Ltd. | Software Engineer | Nov 2008 – Aug 2011

Projects

Independent DevOps Portfolio | Oct 2025 – Present

Independently built and maintained live DevOps projects to demonstrate hands-on experience with Azure DevOps, Kubernetes, Docker, Terraform, PowerShell, security and observability.

Project Evidence: Live applications and public technical documentation are available through the DevOps portfolio. Four private Git repositories containing 190+ commits document the staged implementation of the MHC and ERP projects and can be demonstrated during technical interviews.

My Health Clinic – Self-Hosted Kubernetes Platform | Jan 2026–Present | Live: <https://mhc.sunitdevops.site>

Description:

MHC is a cloud-native healthcare application deployed on a self-hosted Kubernetes platform with automated CI/CD, Infrastructure as Code, secrets management, observability, and multi-environment delivery.

Key Contributions:

- Modernized the application to .NET 6 and Angular, separated the Web and API components using REST communication, and implemented JWT-based authentication and user registration.
- Built a four-node self-hosted Kubernetes cluster with one control-plane and three worker nodes using kubeadm, containerd and Calico.
- Containerized the Web, API and DACPAC deployment components using Docker, published versioned images to Azure Container Registry, and managed Dev, Pre-Prod and Prod deployments using Kustomize.
- Implemented a multi-stage Azure DevOps YAML pipeline for Terraform validation, Docker builds, image publishing, database deployment and Kubernetes deployments, executed through a self-hosted Azure DevOps agent.
- Implemented Terraform with an Azure Storage remote backend and environment-specific state management.
- Integrated HashiCorp Vault AppRole authentication for Kubernetes Secrets and configured Workload Identity Federation for secure Azure access without long-lived client secrets.
- Configured secure public access using Cloudflare Tunnel, NGINX Ingress, cert-manager and Let's Encrypt TLS certificates.
- Implemented monitoring with Prometheus and Grafana and centralized logging with Fluent Bit, Elasticsearch and Kibana.

Tech Stack: Azure DevOps, Terraform, Azure Storage, Docker, Kubernetes, kubeadm, containerd, Calico, Kustomize, Azure Container Registry, Workload Identity Federation, HashiCorp Vault, NGINX Ingress, cert-manager, Prometheus, Grafana, Fluent Bit, Elasticsearch, Kibana, Cloudflare Tunnel, Self-Hosted Azure DevOps Agent, .NET 6, Angular, SQL Server

Employee Recruitment Portal | Oct 2025–Dec 2025 | Live: <https://erp.sunitdevops.site>

Description:

Employee Recruitment Portal is a .NET-based web application for managing job openings, candidate referrals, CV screening, interviews, and job offers. The project demonstrates automated CI/CD deployment to a Windows/IIS environment.

Key Contributions:

- Implemented an Azure DevOps YAML CI/CD pipeline using a self-hosted agent to build, publish and deploy the application and database to a Windows/IIS host.
- Automated IIS website, application pool, deployment-directory and application-binding management using PowerShell.
- Implemented artifact-based application deployment and automated SQL database deployment using DACPAC and SqlPackage.
- Used an Azure DevOps Library Variable Group to manage environment-specific database, deployment-path, IIS and configuration values.
- Used PowerShell Remoting to automate application backup, file transfer, configuration updates, IIS restart and post-deployment validation, and configured public access using Cloudflare DNS and Cloudflare Tunnel.

Tech Stack: Azure DevOps, GitHub, PowerShell, PowerShell Remoting, Self-Hosted Agent, IIS, SQL Server, DACPAC, SqlPackage, Cloudflare Tunnel, .NET

Professional Projects

Encora Innovation Labs | Senior Azure DevOps Engineer | Mar 2025 – Sep 2025 | GoDaddy Client

Project: Ecom Message Processor

Description:

Ecom Message Processor is a Windows service deployed via Chocolatey that retrieves entitlement data and integrates with GDAPI to create or update entitlement records.

Key Contributions:

- Migrated the Ecom Message Processor deployment from Jenkins to GitHub Actions by converting the existing Jenkinsfile into an automated GitHub Actions workflow.

- Integrated PowerShell and Chocolatey commands to automate Windows service installation and upgrades.
- Used PowerShell Remoting to connect to target servers and deploy or update the Ecom Message Processor service.
- Implemented environment-specific deployment workflows for Dev, Test and Prod, with secure management of GitHub secrets.
- Automated the end-to-end deployment process, reducing manual intervention and improving release consistency.

Tech Stack: Jenkins, GitHub Actions, PowerShell, PowerShell Remoting, Chocolatey, Windows Services, .NET Core

Encora Innovation Labs | Senior Azure DevOps Engineer | Dec 2023 – Feb 2025 | GoDaddy Client

Project: GDAPI Migration & CI/CD Modernization (GoDaddy API Integration)

Description:

GDAPI is a set of services that enables developers and partners to manage GoDaddy hosting and domain resources programmatically. It supports operations such as domain purchase, renewal and transfer, DNS management, domain availability, account information, FTP users, database servers and SSL certificates.

Key Contributions:

- Migrated the legacy GDAPI application from .NET Framework to .NET Core using .NET Upgrade Assistant, enabling cross-platform application support.
- Replaced the existing Jenkins pipeline with GitHub Actions and implemented an automated CI/CD workflow.
- Containerized the application by creating Dockerfiles to provide consistent builds and deployments across environments.
- Set up and configured a self-managed Kubernetes cluster for the Development zone and deployed the containerized GDAPI services.
- Deployed and tested the containerized application in Kubernetes and validated API functionality, application integrations and workloads in the Development environment.
- Worked within a multi-zone architecture covering Development and Production, with direct ownership of the Development environment and collaboration with dedicated teams for Production deployments.
- Configured separate Dev, Test and Prod environments in GitHub Actions to support controlled and secure releases.
- Installed and configured GitHub Actions runners to execute build and deployment jobs in a secure, isolated environment.
- Structured the CI/CD workflow into separate build, test and deployment stages to improve maintainability and failure isolation.
- Integrated existing PowerShell automation scripts into GitHub Actions for environment setup and deployment.

Tech Stack: GitHub Actions, Jenkins, Docker, Kubernetes, PowerShell, .NET Upgrade Assistant, ASP.NET Core Web API, .NET Framework, .NET Core, SQL Server

T-Systems ICT India Pvt. Ltd. | Azure DevOps Engineer | Dec 2018 – May 2023 | Royal Dutch Shell Client

Project: Shell Lube Analyst

Description:

Shell Lube Analyst is an enterprise oil condition monitoring platform supporting end-to-end sample processing, reporting and master data management for internal teams and external customers. The cloud-based platform was modernized by Microsoft's Hyderabad engineering team using Azure PaaS services, with ongoing CI/CD, cloud infrastructure, release and production support activities delivered in collaboration with Shell teams.

Key Contributions:

- Designed and implemented Azure DevOps CI/CD pipelines to automate build and release processes across multiple environments.
- Independently provisioned and configured the Staging environment in Azure to support application testing and controlled deployments.
- Provisioned and managed Azure PaaS services, including Azure App Service, Azure Functions, Azure Storage, Azure Service Bus, Azure SQL and Logic Apps, using ARM Templates.
- Automated the cross-subscription migration of an Azure SQL database of approximately 250 GB using PowerShell, reducing more than 15 hours of manual effort.
- Resolved a critical production issue that prevented sample reports from being delivered to customers. Reproduced the failure through direct .NET REST API testing and traced the Azure Functions and Azure Service Bus workflow to identify attachment-size growth during PDF, Excel and Word email generation as the cause of a System.IO.IOException.

- Administered Microsoft Entra ID B2B collaboration and resolved more than 500 external-user authentication and access issues.
- Implemented Point-to-Site VPN connectivity for secure and controlled access to Azure virtual machines.
- Monitored and optimized Azure infrastructure performance using Application Insights and Azure Monitor and managed backup and scaling activities to maintain application availability and reliability.
- Collaborated with development and QA teams to ensure environment readiness and smooth release cycles.
- Troubleshooted and resolved critical cloud-, application- and deployment-related Production issues.
- Transitioned infrastructure provisioning from ARM Templates to modular Terraform after encountering template parameter and variable limitations, improving maintainability and scalability.
- Supported AKS workload deployment and monitoring in coordination with the platform team.
- Developed PowerShell automation using Microsoft Graph API for Entra ID user management, B2B collaboration and access governance activities.

Tech Stack: Azure DevOps, Azure Repos, Terraform, ARM Templates, AKS, Azure App Service, Azure Functions, Azure SQL, Azure Storage, Azure Service Bus, Azure Key Vault, Logic Apps, Microsoft Entra ID, Application Insights, Azure Monitor, PowerShell, Microsoft Graph API, .NET REST API

Earlier Enterprise .NET Development Experience | 2008–2018

Before transitioning to Azure DevOps and cloud engineering, gained approximately **nine years of enterprise .NET development experience** across healthcare, banking, engineering, and financial services domains.

- Developed web applications, Windows services, APIs, database solutions, and integration workflows using C#, ASP.NET MVC, .NET Framework, SQL Server, Entity Framework, SOAP Web Services, TFS, and JavaScript.
- Modernized BMW Financial Services integrations by replacing BizTalk workflows with .NET services and queue-based processing.
- Designed and developed the Johnson Controls Engineering Change Notification application, including database architecture, workflow modules, and legacy-data migration.
- Delivered enterprise application development and production support for Standard Chartered and Pellicid Healthcare.

Recognition:

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- Quarterly awards for the Shell Lube Analyst project – Dec 2019 and Mar 2021.